

# Zhankun Luo

## Machine Learning Theory & Optimization

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### RESEARCH PROFILE

I develop and analyze algorithms for stochastic optimization and statistical learning, with an emphasis on convergence and finite-sample guarantees. My work spans variance reduction and randomized methods, distributed optimization under stochastic constraints, and the dynamics and statistical guarantees of latent-variable learning. My ongoing research investigates oracle-complexity lower and upper bounds for nonconvex optimization under state-dependent heavy-tailed gradient noise.

### EDUCATION

#### Purdue University

*Ph.D. in Electrical and Computer Engineering*

Advisor: Prof. Abolfazl Hashemi; Machine Intelligence & Networked Data Science (MINDS) Group.

**Present**

*West Lafayette, IN*

#### Purdue University Northwest

*M.S. in Electrical and Computer Engineering*

**2021**

*Hammond, IN*

#### Beijing Institute of Technology

*B.S. in Telecommunication Engineering*

**2019**

*Beijing, China*

### SELECTED THEORETICAL CONTRIBUTIONS

#### Variance reduction and randomized algorithms

- › Developed a unified high-probability analysis of stochastic variance-reduced estimators using a new Freedman-type inequality, including mirror-descent guarantees in Banach spaces and improved oracle complexity for expectation-constrained optimization. 📄 [paper](#) 📄 [poster](#) [5] [NeurIPS 2026]
- › Developed unbiased randomized midpoint estimators that correct discretization bias in nonlinear extragradient updates; used antithetic sampling to cancel first-order variance terms and established guarantees for root finding, constrained variational inequalities, and smooth convex-concave games. 📄 [paper](#) [3]
- › Formulated strong antithetic maps and indices to maximize negative dependence while preserving a target law, connecting optimal antithetic coupling to geometric optimal transport for Monte Carlo integration, approximation, and stochastic optimization. 📄 [slides](#) [2]

#### Distributed stochastic minimax optimization with constraints

- › Designed a single-loop, primal-only Softmax-weighted switching-gradient method for worst-client optimization under stochastic constraints; established convergence under full and partial client participation with logarithmic high-probability dependence. 📄 [paper](#) 📄 [poster](#) 🔄 [code](#) [6] [UAI 2026]

#### Dynamics and statistical guarantees for latent-variable models

- › Derived explicit population EM updates across all signal-to-noise regimes; proved that noiseless iterates lie on a cycloid, estimated the super-linear convergence exponent, and improved finite-sample statistical error bounds. 📄 [paper](#) 📄 [poster](#) [9] [ICML 2024]
- › Extended the trajectory analysis to unknown mixing weights and regression parameters, establishing linear-to-quadratic population convergence and arbitrary-initialization finite-sample guarantees. 📄 [paper](#) 🔄 [code](#) [7] [IEEE TIT R&R; revised manuscript submitted]
- › Characterized overspecified EM with jointly estimated regression parameters and mixing weights, proving distinct linear and sublinear regimes and deriving iteration, sample, and statistical-accuracy guarantees in population and finite samples. 📄 [paper](#) 🔄 [code](#) [8] [TMLR 2026]
- › Proved consistency and asymptotic normality with maximum-likelihood covariance for path-integrated score matching in mixed linear regression with unknown mixing weights under suitable diffusion schedules; derived exact fixed-noise decompositions linking the score-matching loss and gradients to EM operators. 📄 [paper](#) [4]

#### Stochastic optimization under state-dependent heavy-tailed noise

- › Investigating oracle-complexity lower bounds and algorithmic upper bounds for nonconvex optimization under state-dependent bounds on gradient-noise moments. 📄 [slides](#) [1]

## PUBLICATIONS & MANUSCRIPTS

- [1] **Z. Luo**, A. Fazla, A. Upadhyay, S. B. Moon, B. Shin, and A. Hashemi. “Stochastic Nonconvex Optimization with Heavy-Tailed Noise: Distance and Suboptimality Dependence.” *Manuscript in preparation*.
- [2] **Z. Luo**, A. Makur, and A. Hashemi. “Generalized Antithetic Variance Reduction: an Optimal Transport Approach.” *Manuscript in preparation*.
- [3] **Z. Luo**, M. B. Sahin, A. Upadhyay, B. Sharif, and A. Hashemi. “RAMPAGE: RAnimized Mid-Point for debiAsed Gradient Extrapolation.” *Preprint*, 2026. Submitted to *International Conference on Artificial Intelligence and Statistics (AISTATS)*. [paper](#)
- [4] **Z. Luo** and A. Hashemi. “Connecting Score Matching, Maximum Likelihood, and Expectation-Maximization in Mixed Linear Regression.” *Preprint*, 2026. Submitted to *Transactions on Machine Learning Research (TMLR)*. [paper](#)
- [5] **Z. Luo**, A. Upadhyay, M. B. Sahin, S. B. Moon, A. Makur, and A. Hashemi. “Unified High-Probability Analysis of Stochastic Variance-Reduced Estimation.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. [paper](#) [poster](#)
- [6] **Z. Luo**, A. Upadhyay, S. B. Moon, and A. Hashemi. “First-Order Softmax Weighted Switching Gradient Method for Distributed Stochastic Minimax Optimization with Stochastic Constraints.” *Uncertainty in Artificial Intelligence (UAI)*, 2026. [paper](#) [poster](#) [code](#)
- [7] **Z. Luo** and A. Hashemi. “Structural Properties, Cycloid Trajectories and Non-Asymptotic Guarantees of EM Algorithm for Mixed Linear Regression.” *Revised manuscript submitted to IEEE Transactions on Information Theory following a revise-and-resubmit decision*. [paper](#) [code](#)
- [8] **Z. Luo** and A. Hashemi. “Characterizing Evolution in Expectation-Maximization Estimates for Overspecified Mixed Linear Regression.” *Transactions on Machine Learning Research (TMLR)*, 2026. [paper](#) [code](#)
- [9] **Z. Luo** and A. Hashemi. “Unveiling the Cycloid Trajectory of EM Iterations in Mixed Linear Regression.” *International Conference on Machine Learning (ICML)*, 2024. [paper](#) [code](#)
- [10] E. Cai, **Z. Luo**, S. Baireddy, J. Guo, C. Yang, and E. J. Delp. “High-Resolution UAV Image Generation for Sorghum Panicle Detection.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2022. [paper](#) [video](#)
- [11] N. J. Walla, **Z. Luo**, B. Chen, Y. Krotov, and C. Q. Zhou. “Smart Ladle: AI-Based Tool for Optimizing Caster Temperature.” *Iron & Steel Technology Conference (AISTech)*, 2021. [paper](#) [poster](#)
- [12] **Z. Luo**. “Structured Light Vision Systems Using a Robust Laser Stripe Segmentation Method.” M.S. thesis, Purdue University, 2021. [paper](#)
- [13] **Z. Luo**, Y. Zhang, and L. Tan. “Multi-Level Random Sample Consensus Method for Improving Structured Light Vision Systems.” *IEEE UEMCON*, 2020. [paper](#)
- [14] Y. Zhang, **Z. Luo**, J. Hou, L. Tan, and X. Guo. “Computer Vision Techniques for Improving Structured Light Vision Systems.” *IEEE EIT*, 2020. [paper](#)

## RESEARCH EXPERIENCE

### Research Assistant, MINDS Group, Purdue University

Aug. 2025 – Present

PI: Prof. Abolfazl Hashemi

West Lafayette, IN

- › Design variance-reduced and randomized methods for stochastic minimax and expectation-constrained optimization, with convergence and oracle-complexity guarantees.
- › Develop federated-optimization algorithms for full and partial client participation under communication and resource constraints within the NSF project [Learning through the Air: Cross-Layer UAV Orchestration for Online Federated Optimization](#).

### Research Assistant, VIPER Laboratory, Purdue University

Aug. 2021 – May 2022

PIs: Prof. Edward J. Delp and Prof. Fengqing M. Zhu

West Lafayette, IN

- › Investigated hierarchy-aware learning for fine-grained food recognition in the [Technology Assisted Dietary Assessment \(TADA\) project](#) and generated labeled UAV imagery with image-to-image GANs for the [PhenoSorg plant-phenotyping project](#).

### Research Assistant, CIVS, Purdue University Northwest

Feb. 2020 – May 2021

PIs: Prof. Chenn Zhou, Prof. Bin Chen, and Prof. Lizhe Tan

Hammond, IN

- › Built DNN and LightGBM models for the [Smart Ladle steel casting-temperature predictor](#) using process data from SQL databases; developed robust multi-camera, multi-laser algorithms for structured-light metrology.

## TEACHING EXPERIENCE

### Teaching Assistant, Purdue University ECE

May 2022 – Aug. 2025

ECE 69500: Optimization for Deep Learning; ECE 20001/20002: EE Fundamentals

West Lafayette, IN

- › **ECE 69500**: Supported instruction in stochastic gradient methods, generalization, and distributed learning; held office hours and graded homework and exams. [slides](#) [video](#)
- › **ECE 20001/20002**: Led office hours and help sessions, graded assessments, coordinated TA activities, and compiled an ECE 20001 practice-problem collection. [problems](#) [video](#)

## SELECTED PRESENTATIONS

- › **First-Order Softmax Weighted Switching Gradient Method for Distributed Stochastic Minimax Optimization with Stochastic Constraints.** Online presentation, UAI 2026, Aug. 2026. [poster](#)
- › **Poster presentation**, Midwest Machine Learning Symposium, Purdue University, Jun. 2026. [event](#)
- › **Strong Antithetic Variates: Theory and Applications.** MINDS Group presentation, Purdue University, Nov. 2025. [slides](#)
- › **High-Resolution UAV Image Generation for Sorghum Panicle Detection.** Oral presentation, Agriculture-Vision Workshop at CVPR 2022, New Orleans, LA, Jun. 19, 2022. [video](#)

## PROFESSIONAL SERVICE

**Journal reviewer:** IEEE Transactions on Signal Processing; Stochastic Systems.

**Conference reviewer:** NeurIPS; ICLR; UAI; AISTATS; AAAI; IEEE ISIT; IEEE ICME; CVIS.

**Memberships:** IEEE Young Professionals; IEEE Signal Processing Society; IEEE Computer Society.

## SELECTED RESEARCH SOFTWARE

|                                           |                                                                                                                                                 |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">SoftmaxSGM</a>                | Implementation of the UAI 2026 Softmax-weighted switching-gradient method.                                                                      |
| <a href="#">em_overspecified_mlr</a>      | Numerical experiments for the TMLR 2026 analysis of overspecified EM.                                                                           |
| <a href="#">cycloid_em_mlr</a>            | Numerical experiments and visualizations for the ICML 2024 cycloid analysis.                                                                    |
| <a href="#">smart_ladle</a>               | LightGBM casting-temperature predictor with SQL integration, tested and deployed at Steel Dynamics' Butler Division. <a href="#">SDI Butler</a> |
| <a href="#">hierarchy_classification</a>  | PyTorch reimplement of hierarchy-aware food classification and embedding.                                                                       |
| <a href="#">Kmeans-tsne-visualization</a> | C++/Qt/OpenGL application for interactive visualization of t-SNE embeddings and k-means iterations.                                             |

## LEADERSHIP, MENTORING & OUTREACH

- › **Graduate Student Mentor, English Training in Engineering (ETIE):** designed machine-learning and computer-vision project frameworks and mentored undergraduate research in sequential prediction and 3D reconstruction.
- › **Coordinator and Host, Science Communication Contest:** coordinated participants and judges and organized a science communication program serving more than 3,600 students.
- › **Team Leader, Mathematical Modeling Contests:** led model development and implementation for national and international competitions, earning a CUMCM Second Prize and an ICM Honorable Mention.

## SELECTED AWARDS & HONORS

|                                                                           |      |
|---------------------------------------------------------------------------|------|
| NeurIPS Conference Travel Grant                                           | 2025 |
| AIST Hunt-Kelly Outstanding Paper Award, third place                      | 2022 |
| AIST Digitalization Applications Technology Best Paper Award              | 2021 |
| Innovative Practice Outstanding Student (top 5%)                          | 2019 |
| National Third Prize, Energy Saving and Emission Reduction Competition    | 2017 |
| Interdisciplinary Contest in Modeling, Honorable Mention                  | 2017 |
| Contemporary Undergraduate Mathematical Contest in Modeling, Second Prize | 2016 |

## TECHNICAL BACKGROUND

**Theory:** stochastic, constrained, minimax, and nonconvex optimization; variance reduction; concentration and martingale inequalities; oracle complexity bounds; non-asymptotic statistics; federated learning; dynamical systems; optimal transport.

**Programming & scientific computing:** Python, Mathematica, MATLAB, C/C++, SQL, Bash, C#, Java; NumPy, SciPy, pandas.

**Machine learning & vision:** PyTorch, TensorFlow, LightGBM, scikit-learn, OpenCV; multi-view geometry, camera calibration.

**Research software & tools:** Linux, Git, Docker, CMake, gdb, Qt, OpenGL, Unity,  $\LaTeX$ .

## SELECTED COURSEWORK

Real Analysis; Linear Algebra with Applications; Introduction to Applied Mathematics; Introduction to Mathematical Statistics  
Optimization for Deep Learning; Statistical Machine Learning; Probabilistic Machine Learning; Reinforcement Learning  
Random Variables and Signals; Estimation Theory; Pattern Recognition and Decision Making  
Model-Based Image and Signal Processing; Digital Signal Processing I; Digital Image Processing I  
Digital Control Systems Analysis and Design; Object-Oriented Design, Analysis and Programming